$$\frac{1}{v_1}-\frac{1}{-15}=\frac{1}{10}; \qquad \frac{1}{v_1}+\frac{1}{15}=\frac{1}{10}$$

$$\frac{1}{v_1}=\frac{1}{10}-\frac{1}{15}=\frac{15-10}{150}=\frac{5}{150}=\frac{1}{30}$$

$v_1 = 30$ cm

First lens forms image 30 cm to the right of the first lens.

Let us find the height of this image.

Equation for magnification is, $m=\frac{h'}{h}=\frac{v_1}{u_1}$

Substituting the values, $\frac{h'}{0.5}=\frac{30}{-15}$

$$h'=0.5\times\frac{30}{-15}=-1 \text{ cm}$$

As the height of the image is negative, the image is inverted and real.

This image acts as object for second lens. The object distance for second lenses, (40–30=10 cm). Hence, u_2 = –10 cm

For the second lens, the lens equation is,

$$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2}$$

Substituting the values,

$$\frac{1}{v_2}-\frac{1}{-10}=\frac{1}{5}; \frac{1}{v_2}+\frac{1}{10}=\frac{1}{5}$$

$$\frac{1}{v_2}=\frac{1}{5}-\frac{1}{10}=\frac{10-5}{50}=\frac{5}{50}=\frac{1}{10}$$

$v_2 = 10$ cm

The final image is formed 10 cm to the right of the second lens.

Let us find the height of the final image. Assume, the final height of the image formed by the second lens is h'' and the height of the object for the second lens is h'.

Equation for magnification is m' for the second lens is, $m'=\frac{h''}{h'}=\frac{v_2}{u_2}$

Substituting the values, $\frac{h''}{-1}=\frac{10}{-10}$

$$h''=(-1)\times\left(\frac{10}{-10}\right)=1 \text{ cm}=10 \text{ mm}$$

As the height of the image is positive, the image is erect and real.

6.6.7 Silvered lenses

If one of the surfaces of a lens is silvered from outside, then such a lens is said to be a silvered lens. A silvered lens is a combination of a lens and a mirror. Light can enter through the transparent front surface of the lens and get reflected by the sliver coated rear surface. Hence, light travels two times through the lens as shown in Figure 6.38.

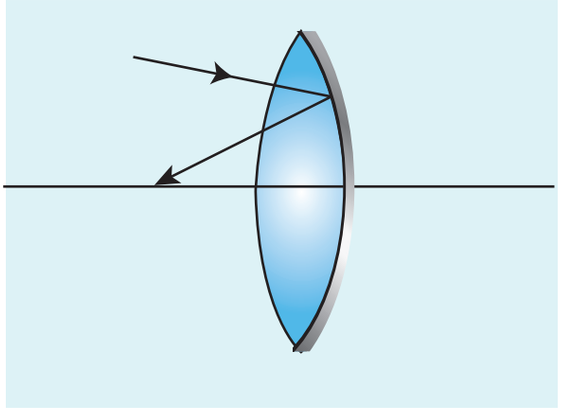


Figure 6.38 Slivered lens

The power P of the silvered lens is,

$$P = P_l + P_m + P_l$$

$$P = 2P_l + P_m \qquad (6.78)$$

Here, P_l is the power of the lens and P_m is the power of the mirror. We know that the power of a lens is the reciprocal of its focal length. But, the power of a mirror is negative of the reciprocal of its focal length. This is because, a concave mirror which has negative focal length is

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